Comprehensive Analysis of the Complex Impacts of the Physical Destruction of Middle Eastern Energy Infrastructure due to the Iran War on the Collapse of the Global Sulfur Supply Chain and Japan's Agricultural and Phosphate Fertilizer Security in 2027

Introduction: Paradigm Shift from Geopolitical Blockade to the Physical Loss of Infrastructure

The historic geopolitical tensions in the Middle East, which dramatically escalated into the "Iran War" in 2026, have caused an unprecedented and fundamental paradigm shift in global energy markets and agricultural supply chains. Initial analyses primarily focused on the "disruption of logistics" accompanying the effective blockade of the Strait of Hormuz, the largest chokepoint connecting the Persian Gulf to the rest of the world. However, the prolongation of the situation and the exchange of armed attacks have shifted the crisis beyond mere logistical route blockades into a more severe and irreversible phase: the "physical destruction" of the densely packed energy production and refining infrastructure in the Gulf region.

As Fatih Birol, Executive Director of the International Energy Agency (IEA), pointed out, approximately 84 energy-related facilities across the Middle East have been damaged, with 34 suffering "serious" or "very serious" physical damage.[1] This destruction does not merely cause an energy crisis characterized by the staggering loss of 13 million barrels per day in crude oil production and 100 billion cubic meters in natural gas supply.[1] It simultaneously means that the production capacity for "sulfur"—a commodity that holds an overwhelming share in global maritime trade—has physically evaporated.

As data from the U.S. Energy Information Administration (EIA) indicates, the Middle East Gulf region accounts for a massive 44% of global sulfur production, a dominant figure that far exceeds its share in hydrocarbon (oil and gas) production.[2] Furthermore, approximately 50% of internationally traded sulfur is supplied from this region.[3] In modern, highly industrialized agriculture, natural gas is indispensable for the production of ammonia (the raw material for nitrogen fertilizers). For the production of phosphate fertilizers (such as MAP, DAP, and TSP), elemental sulfur—the raw material for the sulfuric acid needed to dissolve phosphate

rock—acts as an absolute limiting factor.

This report focuses on the physical destruction of "Sulfur Recovery Units (SRU)" attached to Middle Eastern oil refineries and gas processing plants—rather than the temporary transport bottleneck of the Hormuz blockade—and the structural limits in the supply chain that hinder their restoration. Based on this premise, the report provides a comprehensive, quantitative, and qualitative analysis of how the global evaporation of sulfur supply in the short-to-medium-term timeframe of 2027 will restrict the manufacturing capacity of Morocco, the world's largest phosphate fertilizer exporter, and consequently inflict a fatal blow to Japan's food security and regional agriculture (especially highly profitable crops with high phosphate requirements).

The Full Extent of Infrastructure Destruction in the Middle East Gulf Region and the Direct Blow to Sulfur Production

In past geopolitical crises, such as the drone attacks on Saudi Arabia's Abqaiq and Khurais facilities in 2019, 5.7 million barrels per day of crude processing capacity was temporarily lost, but Saudi Aramco managed to restore production capacity within weeks.[4] However, the ongoing situation is entirely different from any past instance in terms of the scale of the attacks, the breadth of the targets, and the physical destructive power of the weapons used.

Damage to Major Energy Infrastructure and Loss of Sulfur Recovery Capacity

Following a series of retaliatory attacks between March and April 2026, numerous oil refineries, gas processing plants, and export terminals across the Middle East sustained catastrophic damage.[7] Since sulfur is produced as a byproduct in the refining process of high-sulfur (sour) crude oil and the desulfurization process of sour natural gas, the shutdown of these plants immediately translates to a halt in sulfur supply to the global market. Below is a summary of the major damaged facilities directly linked to the sulfur supply chain and their impacts.

Country	Target Facility/Region	Damage Status and Impact on Sulfur Production/Market
United Arab Emirates (UAE)	Habshan	Largest domestic natural gas processing plant. Operations suspended due to attack and fire. Fatal blow to Abu Dhabi National Oil Company (ADNOC)'s

		core sulfur recovery capacity from sour gas.[7]
United Arab Emirates (UAE)	Ruwais	One of the world's largest oil refineries. Suffered multiple fires from air-defense interception debris, forcing a long-term shutdown beyond precautionary measures.[7]
Saudi Arabia	Satorp, Riyadh	Unit shutdown at Satorp, a 460,000 barrel-per-day joint venture refinery between Aramco and TotalEnergies. A 120,000 barrel-per-day plant in Riyadh was also attacked, reducing refining capacity.[7]
Saudi Arabia	Ras Tanura	One of the largest crude processing plants with a 550,000 barrel-per-day capacity. Temporarily halted due to drone attacks. Although reportedly restarted, peripheral equipment damage prevents stable operation.[7]
Qatar	Ras Laffan	LNG facilities and a Gas-to-Liquids (GTL) plant suffered extensive fire damage from missile attacks, declaring force majeure.[7] A major supply source for the global spot sulfur market evaporated.[9]
Kuwait	Mina Al-Ahmadi	Fires in operational units at the refinery. Continuous drone and missile attacks

		severely degraded refining capacity and the accompanying sulfur recovery functions.[7]
Iran	South Pars	Processing units at the giant gas field were damaged by Israeli attacks. Iran's gas supply network and desulfurization processes were disrupted, affecting domestic steel mills and other industries.[7]

Saudi Arabia (annual production approx. 4.5 million tons, design capacity over 6 million tons), the UAE (approx. 3.5 million tons), and Qatar are the "Big Three" sulfur suppliers to the global market.[9] With the central infrastructure of these countries simultaneously suffering physical damage, a sulfur supply capacity of 10 million tons annually is on the verge of disappearing from the global market. Furthermore, over 200 crude oil and petroleum product tankers and 10 LNG tankers are stranded in the Strait of Hormuz, meaning that even the small amount of sulfur produced remains entirely cut off from the market.[1]

The Vulnerability of the Claus Process and Mechanisms of Physical Damage

The IEA's prediction that "even if the blockade of the Strait of Hormuz were lifted, it would take a minimum of two years or more for production levels to recover" [1] is not a pessimistic exaggeration. The reason lies in the technical characteristics of the "Sulfur Recovery Unit (SRU)," a highly advanced chemical plant used to recover sulfur.

Between 90% and 95% of the world's sulfur is recovered using a method called the "Claus Process".[11] Toxic hydrogen sulfide (H2S), separated from natural gas and crude oil refining processes via amine absorption towers, is sent to the SRU and converted into elemental sulfur through the following chemical reactions.

The overall reaction equation is as follows:

$$2H_2S + O_2 \rightarrow 2S + 2H_2O$$

This advanced process mainly consists of two stages. The first is the thermal step, where approximately one-third of the H2S reacts with atmospheric oxygen in a combustion furnace to produce sulfur dioxide (SO2).[11]

$$2H_2S + 3O_2 \rightarrow 2SO_2 + 2H_2O + heat$$

This combustion furnace operates in an extremely harsh, high-temperature environment ranging from 980°C to 1540°C (1800°F to 2800°F), and the immense heat generated is recovered as high-pressure steam in a subsequent waste heat boiler.[11]

The second stage is the catalytic step, where the generated SO2 reacts with the remaining H2S in a catalytic reactor to recover liquid sulfur.[11]

$$4H_2S + 2SO_2 \rightarrow 3S_2 + 4H_2O$$

Not only do direct kinetic attacks by missiles and drones damage these facilities, but sudden power losses caused by the destruction of peripheral infrastructure also inflict fatal damage on the SRU. As seen in accident analyses at North American natural gas plants, if an operating SRU suddenly shuts down due to a power outage, the massive tubesheets and refractory bricks inside the reaction furnace and waste heat boiler experience rapid thermal stress, leading to warping, cracking, and thermal damage (creep and graphitization).[14]

When custom-built, massive pressure vessels designed to withstand such high-temperature, high-pressure, and highly corrosive environments are physically destroyed or thermally deformed, they cannot be fixed with makeshift repairs like simply reconnecting pipes. Advanced welding techniques and a post-weld "Stress-Relieving" process are required to remove internal stresses. If there is even a slight error in execution, the tubesheet will warp again, necessitating a complete remanufacture—a highly complex and difficult repair task.[14]

Structural Barriers in Heavy Industry and Chemical Supply Chains Hindering Recovery

In addition to the physical damage to infrastructure, restoration projects for oil refineries and gas processing plants face a second wall: bottlenecks in the global heavy industry and chemical supply chains. This is the primary reason why recovery will take multiple years.

The Trap of Long-Lead Items and Construction Limitations

In large-scale plant repairs (turnarounds or reconstructions), the procurement lead time for replacement parts determines the project's critical path. Core equipment such as aircraft-sized pressure vessels, heat exchangers, slide valves, compressors, and turbines are classified as "Long-Lead Items".[16]

These items are not off-the-shelf products stored in warehouses; they are entirely fabricated to order. From formulating design specifications to procuring special steel, forging, precision machining, welding, inspection, and shipping, it typically takes 40 to 50 weeks (about a year) even during peacetime.[16]

Furthermore, the current global supply chain for construction materials and heavy industry is suffering from insurmountable delays caused by structural disruptions post-pandemic, severe port congestion in Asia and the U.S., and a global container shortage.[18] With multiple Middle Eastern hubs destroyed simultaneously, orders are flooding the limited number of global plant engineering firms (EPC contractors), triggering fierce competition for production slots. The cost of Non-Productive Time (NPT) caused by delayed equipment arrivals is estimated to reach $125,000 per hour even for mid-sized terminals, financially straining the projects.[19] As a result, the delivery of pressure vessels and waste heat boilers required to rebuild SRUs cannot be confirmed until one and a half to two years after placing an order.

Shrinking Inorganic Catalyst Market and the Retreat of Western Chemical Manufacturers

Compounding the hardware constraints is the extreme shortage of "inorganic catalysts" required to actually run the processes. To achieve a sulfur recovery rate of over 99.9% in the Claus process and the subsequent Tail Gas Treatment Unit (TGTU), specialized activated alumina, titania, or high-performance CoMo (cobalt-molybdenum) and NiMo (nickel-molybdenum) based catalysts are indispensable.[20]

The global inorganic catalyst market is oligopolized by a few chemical and materials manufacturers, such as BASF, Axens, Haldor Topsoe, Johnson Matthey, Clariant, and W. R. Grace.[21] However, many of these Western chemical giants implemented large-scale restructuring and capacity reductions between 2024 and 2026 due to soaring energy prices and sluggish global demand. For example, BASF cut 4,800 employees globally, and Dow Chemical announced a $2 billion cost-reduction plan including 4,500 job cuts.[24] Companies like Solvay and LyondellBasell also proceeded with plant closures and divestments.[24] Additionally, Haldor Topsoe was forced to raise prices across its catalyst product line due to surging raw material and energy costs.[25]

Just as major global catalyst manufacturers were retreating from non-core businesses and reducing production capacity to its absolute limits, a sudden massive demand for "fresh catalyst charges" to rebuild Middle Eastern SRUs emerged. The prolonged lead times and tight supply of catalyst production create a second fatal bottleneck, meaning that even if hardware like pressure vessels were completed, the plants could not be brought to life.[21]

Dysfunction of Alternative Sulfur Supply Chains and Fierce Resource Competition with Other Industries

The disappearance of millions of tons of sulfur annually from the Middle East has triggered a chain reaction of price spikes in global commodity markets and a desperate scramble for alternative suppliers. The misfortune for the agricultural sector is that sulfur is not merely a fertilizer raw material; it is an indispensable substance supporting modern, high-tech industries across the board.

Limitations of Alternative Suppliers (Canada, Kazakhstan, US, Venezuela)

Potential alternative suppliers to offset the Middle East's production loss include Canada, Kazakhstan, the U.S., and Venezuela, but each faces inherent constraints, making short-term substitution extremely difficult.

Canada is one of the world's largest sulfur exporters, producing massive amounts from the processing of oil sands and sour gas in Alberta, primarily exporting from Pacific ports to Asian fertilizer manufacturers.[2] However, the existing rail and export infrastructure connecting inland Canada to its ports has already reached maximum capacity. There is absolutely no logistical leeway to quickly compensate for the shortfall from the Middle East (44% of the global total).[2]

Kazakhstan and Uzbekistan in Central Asia have sulfur production potential driven by energy development led by companies like BP, but they face geopolitical transport barriers unique to landlocked countries. Due to the closure of the Iranian border and stagnant truck transport in the Caucasus (e.g., Azerbaijan), thousands of cargo trucks are stranded, rendering the route completely dysfunctional as a stable supply chain for the global market.[2] Furthermore, while Venezuela generates large amounts of sulfur during the refining of its ultra-heavy, high-sulfur crude oil from the Orinoco Belt, its production and export capabilities remain in a state of collapse due to years of economic sanctions and deteriorating infrastructure.[2]

Competition with Semiconductor and EV Industries and the Subordination of Agriculture

As the supply pie dramatically shrinks, the fertilizer industry is embroiled in fierce competition for sulfur against other high-profit industries. While 60% to 70% of global sulfuric acid is used for fertilizers, the industrial demand making up the remainder is seizing pricing power.[2]

For instance, ultra-pure sulfuric acid used in semiconductor manufacturing is essential for wafer cleaning and printed circuit board production. Companies like Taiwan's TSMC, which produces 90% of the world's advanced chips, face a "double whammy" of depending on the Middle East for both LNG (Qatar supplies 30% of Taiwan's LNG) and sulfuric acid, and are moving to secure alternative raw materials regardless of the premium required.[2]

Additionally, sulfuric acid is a core reagent in the extraction (hydrometallurgy) of essential metals for electric vehicle (EV) batteries, such as copper, cobalt, nickel, and zinc.[2] A typical example is Indonesia's High-Pressure Acid Leach (HPAL) nickel smelting projects; Indonesia had previously increased its sulfur imports from Saudi Arabia and Qatar by over 40% year-on-year.[9] Faced with the overwhelming purchasing power of the technology and EV industries, fertilizer manufacturers—who operate on a low-margin, high-volume business model—are inevitably subordinated in the competition to procure raw materials.

The "Sulfur Trap" of Morocco's OCP (World's Largest

Phosphate Fertilizer Supplier) and its Strategic Shift

The biggest victim of this global sulfur shock and resource competition—and the entity holding the fate of the global agricultural supply chain in 2027—is Morocco's state-owned enterprise, OCP Group (Office Chérifien des Phosphates), which controls approximately 70% of the world's phosphate rock reserves and is the world's largest exporter of phosphate fertilizers.

OCP boasted overwhelming supply capacity in 2024, exporting 4.4 million tons of DAP, 2.71 million tons of MAP, and 2.98 million tons of TSP.[26] However, Morocco lacks domestic sulfur resources and relies almost entirely on imports for the sulfur needed to produce sulfuric acid, the linchpin of fertilizer manufacturing. According to 2025 data, OCP depended on the Middle East (UAE, Saudi Arabia, Qatar, etc.) for a staggering 52% of its millions of tons of annual sulfur imports.[26]

Loss of Production Capacity due to the Sulfur Trap

The blockade of the Strait of Hormuz and the physical destruction of Gulf infrastructure abruptly severed this 52% supply route. Compounded by operational chaos at domestic Moroccan ports due to severe weather in early 2026 [26], OCP has fallen into a severe "Sulfur Trap." Even though its manufacturing plants and port facilities at Jorf Lasfar on the Atlantic coast are undamaged, its advanced sulfur combustion and sulfuric acid extraction plants cannot operate without incoming elemental sulfur.

Under these physical constraints, OCP took the unprecedented step of cutting up to 30% of its production capacity for the second quarter and accelerating its scheduled maintenance.[26] For a company exporting roughly 10 million tons of phosphate fertilizer annually, a 30% loss of production capacity means that about 3 million tons of fertilizer will evaporate from the international market, dealing a fatal supply shock globally.

Shift to the European Market and Lobbying for Cadmium Deregulation

What requires further attention is OCP's "strategic shift" in response to the crisis. Amid the adversity of soaring sulfur prices and reduced production capacity, OCP raised $1.5 billion in international hybrid bonds and deployed a strategy to divert its limited products to the most profitable and strategically important markets.[28]

Specifically, OCP launched intense lobbying efforts targeting the European Union (EU) to ease the bloc's current Fertilisers Regulation.[28] In 2019, the EU introduced strict regulations capping cadmium levels in fertilizers at 60mg/kg for environmental and health protection. Because Moroccan phosphate rock contains up to 80mg/kg of cadmium, this regulation had caused OCP's share in the EU market to drop from 32% in 2018 to 19% currently.[28]

Leveraging concerns over global supply shortages and food security stemming from the Hormuz crisis, OCP is actively attempting to have the EU's environmental regulations relaxed through diplomatic negotiations involving Moroccan Foreign Minister Nasser Bourita and EU High Representative Josep Borrell (or Kaja Kallas), aiming to recapture market share in the

high-priced European market.[28]

This "shift to premium markets" (Europe and North America) and "priority allocation to strategic mega-markets" (like India and Brazil) spells fatal consequences for the Japanese market. With total production volume reduced due to the sulfur shortage, the remaining production quotas will be preferentially allocated to Europe, which pays a premium. Consequently, supplies bound for Japan—which is geographically distant and generally has weaker negotiating power—will face a structural downward pressure.

Protectionism in Major Fertilizer-Producing Countries and the Isolation of the Japanese Market

In addition to physical supply constraints, the nature of fertilizer as a strategic commodity is accelerating "Resource Nationalism" across various countries, completely isolating net-importing countries like Japan from the market.

US Domestic Prioritization and Export Market Distortion

The United States is one of the world's leading producers of phosphate rock and, under normal circumstances, could act as a potent alternative supplier during a global shortage. However, domestic circumstances in the U.S. do not permit this. In the first quarter of 2026, Mosaic, a major U.S. fertilizer manufacturer, exported roughly 201,000 tons of DAP and MAP accumulated in New Orleans (Nola) to international markets outside of Canada, fearing an oversupply during the spring demand season.[29]

This massive export volume resulted from pursuing higher profits by capitalizing on the soaring global prices caused by the Middle East crisis. However, this suddenly tightened fertilizer supplies within the U.S., pushing domestic prices to their highest levels since 2022.[29] Facing skyrocketing farming costs, U.S. farmers and politicians reacted with fierce opposition, driving domestic protectionist pressures on export regulations to the breaking point.[29] It is only a matter of time before the U.S. prioritizes domestic food security and places fertilizers under de facto strategic export controls using the Defense Production Act (DPA) or similar powers, completely eliminating its function as an export buffer for Japan.

Export Controls by China and Russia

China, once the world's largest supplier exporting over 10 million tons of phosphate fertilizer annually, has made suppressing domestic food inflation and securing supply for spring planting its absolute priority, maintaining an effective "total export ban." It has become common practice for local customs offices to refuse export declarations for agricultural phosphate fertilizers and hold up clearance.

Furthermore, Russia, another global fertilizer powerhouse, continues its export quota system for specific fertilizers to maintain reasonable prices for its domestic spring sowing campaign. Additionally, Russian fertilizer, virtually shut out from Western markets due to sanctions, is

strategically diverted to friendly nations centered around the BRICS bloc. It is geopolitically impossible for Japan, which belongs to the Western bloc, to secure alternative procurements from China or Russia in 2027.

Limits of Japan's Policy Responses: Disconnect Between Financial Injection and Physical Shortages

With the cutoff of import routes viewed as a certainty, the question of how to protect Japan's agricultural base is critical. Japan relies on imports for nearly all of its three major fertilizer elements, and overcoming this vulnerability is a pressing national security issue.

The Dilemma of the Government's Emergency Response and Stockpile Releases

The Japanese government reacted swiftly to the outbreak of the Middle East crisis. The Ministry of Agriculture, Forestry and Fisheries (MAFF) and the Ministry of Economy, Trade and Industry (METI) implemented measures to release 15 days' worth of private oil reserves and an additional month's worth of national reserves to curb market panic.[30] Furthermore, a 280 billion yen government fund was activated to 100% subsidize petroleum product costs exceeding specific prices, with an additional 800 billion yen injected from the supplementary budget to maintain the supply chain.[30] Agricultural organizations like JA Zenchu also expressed strong vigilance regarding the long-term surge in fuel and fertilizer raw material prices, working in tandem with the government to secure supplies for the spring planting season.[30]

However, in the phosphate fertilizer crisis, financial subsidies and post-facto price compensation mechanisms offer no fundamental solution. The issue is not that "prices are high," but rather that overseas export quotas are closed, and physically, "there is no fertilizer in the market."

Phosphorus Recovery Technology from Sewage Sludge and the "Energy Trap"

As a trump card to break free from import dependency, the government and industry are strongly promoting the recovery of phosphorus from domestic unutilized resources, notably "Sewage Sludge" and livestock compost. Based on the "Strategy for Sustainable Food Systems, MIDORI" and the "Policy Outline to Strengthen Food Security," the government has set an ambitious target to double the fertilizer utilization of sewage sludge and compost by 2030.[31]

However, in the 2027 paradigm under the Iran War, this scheme faces a fatal dilemma known as the "Energy Trap." While sewage sludge contains valuable phosphorus, it also contains harmful heavy metals and pathogens. To completely detoxify and render these insoluble, an ultra-high-temperature thermal treatment/melting process (such as slagging) is required.

Sustaining this ultra-high-temperature process consumes massive amounts of thermal energy (heavy oil or city gas). The Middle East conflict did not just take away sulfur; it also drove up

global crude oil prices. Despite government subsidies, the prolonged high fuel prices exponentially drive up the running costs of sewage sludge melting plants. As a result, an "economic inversion phenomenon" occurs where the marginal cost of producing recovered phosphorus fertilizer far exceeds the already skyrocketed price of imported chemical fertilizers. Municipalities and businesses can no longer maintain economic viability, forcing them to lower plant utilization rates or simply discard the sludge as incineration ash.

Micro-level Ripple Effects on Local Agricultural Economies: A Case Study of High-Profit Crops in Tokushima Prefecture

The structural shortage of phosphate fertilizers impacts Japanese agriculture far beyond just lowering the calorie-based food self-sufficiency rate; it fundamentally destroys the production base of "high-value-added crops and specialty products" that support local agricultural economies.

In plant physiology, phosphoric acid (P2O5) is known as the "fruiting fertilizer" (migoe). It is a component of ATP, the intracellular energy transfer molecule, and is essential for cell division, root elongation, and flowering/fruiting/ripening. While nitrogen fertilizer can be somewhat substituted with organic matter, a deficiency of available phosphoric acid in the soil has no alternative and directly causes delayed initial growth and significant quality degradation. Here, we analyze the micro-level impacts using the agriculture of Tokushima Prefecture in Shikoku as a model. Tokushima leverages the fertile soil of the Yoshino River basin and a mild climate to cultivate numerous distinctive high-profit crops.[32]

Poor Hypertrophy of Lotus Root and Phosphorus Insolubilization in Soil

Tokushima Prefecture is one of the nation's leading producers of lotus roots, cultivating varieties like "Bichu" and "Daruma".[32] Lotus roots grown in the clay-rich soil of the lower Yoshino River are highly prized in the market for their white color and their balance of fluffiness and crispiness.

Lotus roots are crops that enlarge their underground stems in mud. In the flooded cultivation environments typical for aquatic plants, soil phosphorus tends to bind with iron and aluminum, becoming insoluble and hard for plants to absorb. Therefore, growers conventionally apply ample amounts of phosphate fertilizer as basal and top dressing to ensure phosphorus absorption during the hypertrophy period. If the supply of phosphate fertilizer is cut off in 2027, cell division will stagnate, inevitably resulting in a high incidence of non-standard products due to poor rhizome enlargement and a devastating drop in yields.

Quality Degradation of Naruto Kintoki (Sweet Potatoes)

The "Naruto Kintoki" sweet potato, a flagship brand of Tokushima Prefecture, is grown in sandy

soils rich in minerals from the Naruto Strait, characterized by its bright red color and chestnut-like sweetness.[32] While sweet potatoes are generally said to grow even in poor soil, producing high-quality potatoes stably requires a strict fertilization design that moderates nitrogen while heavily applying potassium and "phosphorus."

A lack of phosphorus impedes the process whereby photosynthetic products (starch) generated in the leaves are translocated to the tuberous roots (potatoes). As a result, not only does root hypertrophy worsen, but the signature sweetness and fluffy texture are compromised, dealing a fatal blow to the brand value cultivated over many years.

Alternate Bearing and Tree Vigor Decline in Sudachi and Kito Yuzu

The impacts are equally severe for acid citrus fruits, such as Sudachi, for which Tokushima Prefecture holds nearly a 100% national share, and Yuzu (like Kito Yuzu), for which it ranks second in production.[32] In fruit tree cultivation, massive amounts of phosphorus are consumed during flower bud formation, flowering, and fruit setting/enlargement.

In years with fertilizer shortages, plants exhibit physiological defense mechanisms such as poor flowering and physiological fruit drop, drastically intensifying alternate bearing (the phenomenon of alternating bumper and poor harvests). In particular, the depletion of stored nutrients (phosphorus) that support early spring budding and flowering leads to a decline in tree vigor over several years. The soaring prices and physical shortage of fertilizer sap the motivation of these Tokushima farmers, triggering an exodus from farming in an already aging regional agricultural sector.

Quantitative Estimation Model of Japan's Phosphate Fertilizer Secured Volume in 2027

Based on the multifaceted qualitative analyses, geopolitical factors, and physical constraints detailed above, we estimate the supply volume of phosphate fertilizer (P2O5 equivalent) that Japan can actually secure in 2027 using a mathematical framework.

Let D_{base} (100% baseline) be Japan's annual import requirement/demand for phosphate fertilizer during peacetime (before the Middle East crisis and China's total embargo). The total secured volume for Japan in 2027, S_{2027}, is the sum of supplies from each procurement source.

$$S_{2027} = I_{China} + I_{Morocco} + I_{Others} + R_{domestic}$$

The estimated securement rate matrix for 2027 is as follows:

Procurement	Baseline	Estimated	Effective	Mechanisms

Source	Dependency	Securement Rate (vs. Baseline)	Contribution Rate (Total)	of Reduction and Constraint
Imports (China)	40.0%	0.0%	0.0%	Complete halt of exports due to domestic inflation control and customs clearance suspension.
Imports (Morocco, etc.)	35.0%	21.0%	7.35%	30% production cut due to the sulfur trap.[26] Strategic priority allocation to the European market amid EU cadmium deregulation negotiations.[28]
Imports (US, Russia, Others)	5.0%	15.0%	0.75%	Domestic prioritization due to backlash from US farmers [29], sanctions against Russia, and Canada's transport limitations.[2]
Domestic Recovery (Sludge, etc.)	20.0%	120.0%	24.0%	Expansion efforts continue toward

				resource utilization targets [31], but remain at a slight increase due to operational limits on melting furnaces caused by soaring crude oil prices (Energy Trap).
Total Secured Rate	**100.0%**	-	**Approx. 32.1%**	(A fatal shortage of approx. 67.9% against baseline requirements)

As this model clearly demonstrates, even if we assume an extremely optimistic utilization of domestic resources and continued (albeit drastically reduced) imports from Morocco, Japan in 2027 will face a desperate supply shock, securing only about 32% of its peacetime phosphate fertilizer requirements.

Conclusion and Recommendations for Mid-to-Long-Term Survival Strategies

The 2027 scenario derived from this analysis warns of an unprecedented crisis for Japan's agricultural supply chain. The destruction of Middle Eastern infrastructure triggered by the Iran War goes beyond a mere disruption of global energy supplies; it structurally destroyed the phosphate fertilizer manufacturing capacity of countries like Morocco through the invisible chemical mechanism of "sulfur's physical evaporation." Furthermore, tight catalyst supply chains and the scramble for sulfuric acid against the semiconductor and EV industries are hopelessly delaying recovery.

This shock has ignited survival instincts across nations, inducing complete export bans by China and inward-looking policies in the US, and compelling Morocco to pivot to the European market, thereby completely isolating net-importing countries like Japan. Additionally, the technology for recovering phosphorus from sewage sludge—considered Japan's "last resort"—is caught in a dilemma where it cannot fully exert its potential due to the ripple effects

of the energy crisis and skyrocketing crude oil prices.

Against the forecast that Japan's secured volume of phosphate fertilizer in 2027 will plummet to less than a third of peacetime levels (approx. 32%), conventional market-based approaches like "buying at higher prices" or traditional subsidy systems [30] will be rendered dysfunctional. To mitigate this fatal supply-demand gap and maintain national food security, the following drastic and discontinuous measures are required:

1. **Separation of Energy Subsidies for Fertilizer Plants as "Security Infrastructure":** The massive heavy oil and gas costs required for recovering phosphorus from sewage sludge (slagging processes) must be detached from the independent accounting frameworks of agriculture and sewage businesses, and directly subsidized from a "National Security Budget." Establishing a legal and financial framework where the national treasury fully covers the energy costs for detoxifying heavy metals, thereby running existing melting plants at full capacity regardless of economic profitability, is an urgent priority.
2. **Deployment of "Non-Price" Strategic Barter Diplomacy with Morocco:** While the EU attempts to secure the market through deregulation premiums [28], Japan must break away from simple price competition. Japan needs to immediately conclude long-term supply agreements based on state-level infrastructure barter transactions—such as offering its advanced water treatment technology, renewable energy technology, or hydrogen technology to keep Morocco's phosphate plants running—to politically lock in supply quotas for 2028 and beyond.
3. **Emergency Implementation of Fertilizer Reduction Technologies and Alternative Farming Methods:**
   To survive 2027 when fertilizers physically do not exist, a nationwide forced transition program to extreme fertilizer-reduction models must be prepared. This includes utilizing specialized soil microbes (like mycorrhizal fungi) to dissolve the insoluble "accumulated phosphorus" that has built up in Japanese farmlands over decades, and implementing precision soil diagnostics. Specifically, to protect local high-profit crops like Tokushima's specialties, a logistics network must be established to strategically and preferentially allocate the limited phosphorus resources.

The black smoke rising from Middle Eastern infrastructure has brought to light the fact that modern industrialized agriculture rests on an extremely fragile foundation of "fossil fuels and their accompanying chemical byproducts." As the weaponization of resources and the fragmentation of supply chains progress irreversibly, the phosphate fertilizer shortage Japan faces in 2027 is not a transient price shock, but a structural threat questioning national survival and the very existence of its agricultural infrastructure.

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